

Aayan Atif

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Technical Skills

Languages: C, C++, Python, Java, JavaScript, TypeScript, HTML, CSS, SQL

Frameworks / Platforms: Node.js, React, Express.js, FastAPI, MongoDB, Git, PyTorch npm, AWS, Azure, Docker, Git

Concepts: REST API Design, Data Structures & Algorithms, Database Design, Performance Optimization, Test Automation, Authentication & Authorization, Concurrent Programming, API Security, CI/CD, System Design, Machine Learning, DevOps

Projects

Ts-Rx: Self-Healing Dashboard for Tailscale Networks | *Python, FastAPI, React, TypeScript* HackCanada 2026

- Improved reliability across up to **100** self-hosted **Tailscale** services by building an **AI-powered monitoring agent** that used **Datadog** telemetry and **Gemini** diagnostics with service-specific markdown knowledge bases to detect failures and generate targeted recovery solutions.
- Enhanced usability for self-hosted infrastructure management by injecting a custom **"Doctor" tab** into the **Tailscale** interface, allowing teams to monitor service health, review diagnostics, and access **AI-generated** recovery suggestions from a centralized dashboard.
- Built a self-improving remediation pipeline that saved **Gemini**-generated fixes into service-specific **markdown knowledge bases**, enabling faster resolution of recurring failures through reusable incident history and accumulated service context.

a11y.sense: AI Accessibility QA Sandbox | *Next.js, React, Three.js, Gemini AI, Playwright* GenAIGenesis 2026

- Improved accessibility issue detection across live websites by developing a **Playwright** and **Gemini**-powered auditing pipeline that analyzed DOM structure, screenshots, and element coordinates through **5 parallel Gemini** accessibility audits spanning WCAG principles to generate actionable fixes and precise developer handoff reports.
- Increased accessibility audit interpretability by building an interactive **Three.js**-based **3D skeleton dashboard** that visualized WCAG violations across **5 disability-focused body systems** and linked each issue to affected users, severity, fixes, and raw HTML elements.
- Reduced friction between accessibility audits and implementation by building one-click **Markdown** and **PDF** export systems that converted **Gemini**-generated WCAG findings into developer-ready reports with CSS selectors, affected HTML elements, remediation steps, and prioritization metadata.

Scroll4Good: Charity Impact Platform | *JavaScript, HTML/CSS, Node.js, Express.js, MongoDB* UmmahHacks 2026

- Increased access to charitable support by building a **TikTok**-style short-form video platform that connected users with individuals and communities in need through video feeds, enabling direct **Stripe** donation workflows and achieving **15+ completed donations** across **20+ early users**.
- Strengthened backend security and intelligence by implementing **JWT** authentication, role-based access control, rate limiting, **Zod** validation, **Stripe** webhooks, and a **Gemini**-based AI risk scoring pipeline to secure user actions and prioritize donation requests based on structured severity signals.
- Increased donor retention by implementing a gamified **tree-to-forest progression system** that visually represented donation impact through persistent seed-to-tree growth mechanics tied to user contribution history.

Education

Wilfrid Laurier University

Expected May 2027

Bachelor of Science in Computer Science

Relevant Coursework: Data Structures and Algorithms I and II, Discrete Math, OOP, Microprocessors, Linear Algebra

Experience

CKSS Robotics Club

October 2022 – July 2023

Robotics Team Member

Milton, ON

- Contributed to **winning 1st place at the Ontario Regional FIRST Robotics Competition**, demonstrating innovative problem-solving, effective teamwork, leadership, creativity, and technical excellence.
- Collaborated with a **20-member** team to design, program, and construct a competitive robot for the FIRST Robotics Competition while honing technical and leadership skills.
- Diagnosed and resolved 30+ technical challenges during design and testing phases to ensure peak robot performance, reliability, efficiency, safety, and consistent functionality.